The Conway graph srg(99, 14, 1, 2) has no automorphism of order 3

An obstruction from discriminants

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Candidate proof, not yet reviewed by an independent reader.
Everything has been verified in Lean 4 / Mathlib (see §6), except Corollary 8, which relies on published results.

Abstract

Let Γ be a strongly regular graph with parameters $(99, 14, 1, 2)$ and let g be a fixed-point-free automorphism of order 3. The action of g gives two 33×33 matrices: the quotient Q with integer entries and a Hermitian matrix M with entries in the Eisenstein integers $\mathbb{Z}[\omega]$, congruent modulo $1 - \omega$. We show that such a pair cannot exist: the lattices of eigenvectors with eigenvalue 3 for Q and M have equal reductions modulo 3, hence determinants congruent modulo 3. Yet the first is 2 modulo 3, because of the factor 11 in $\langle j, j\rangle = 33$, whereas the second is 1, since only the prime 7 occurs there. The case of a fixed triangle is handled in the same way, once one observes that the fixed-point coordinates are divisible by 3. An elementary counting argument (which reproves a result of Makhnev and Minakova) shows that an automorphism of order 3 has at most three fixed points; consequently, Γ has no automorphism of order 3. The fixed-point-free case was the only case of prime order left open by the computer search of Behbahani and Lam; together with published results, this implies that the automorphism group of Γ is trivial or has order 2. The proof uses elementary linear algebra over $\mathbb{Z}$ and $\mathbb{Z}[\omega]$, without computer calculation; it is fully verified in Lean 4, from the definition of a strongly regular graph, without any external input.

1 Statements

Let A be the adjacency matrix of $\Gamma = \mathrm{srg}(99, 14, 1, 2)$; $A^2 = 14I + A + 2(J - I - A)$, or

$$A^2 + A = 12I + 2J, \tag{1}$$

where J is the all-ones matrix. Write j for the all-ones vector, $\omega = e^{2i\pi/3}$, and $\pi = 1 - \omega$.

Theorem 1. *Γ has no fixed-point-free automorphism of order 3.*

Proposition 2. *Γ has no automorphism of order 3 with exactly three fixed points (which would form a triangle).*

Corollary 3. *Γ has no automorphism of order 3.*

Proof. Let $g \neq 1$ with $g^3 = 1$. By Lemma 7 (§5), g has at most three fixed points. Their number is congruent to $99 \equiv 0 \pmod 3$, since the other vertices form orbits of size 3: it is therefore 0 or 3, and Theorem 1 and Proposition 2 give the conclusion. (Lemma 7 is due to Makhnev and Minakova [MM04], in the form "the fixed points form either the empty set or a triangle"; the case with fixed points is also ruled out by a computer search in [B09, Theorem 4.14] and [BL11].) □

Context. The existence of an $\mathrm{srg}(99, 14, 1, 2)$ is an open problem; it is one of Conway's five \$1,000 problems [Co]. See [W84] for an early study of this graph and [BVM22] for the general setting. For the group $G = \mathrm{Aut}(\Gamma)$, the following was known.

- Makhnev and Minakova [MM04], using character theory: $|G|$ divides $2 \cdot 3^3 \cdot 7 \cdot 11$, and an automorphism of order 3 fixes either no vertices or the three vertices of a triangle.
- Behbahani and Lam [B09, BL11], by a computer search (orbit matrices): only 2 and 3 can divide $|G|$, and an automorphism of order 3 has no fixed points [B09, Theorem 4.14]. The case of a fixed-point-free automorphism of order 3 is the only one left undecided by that computation [B09, Table 21].
- Crnković and Maksimović [CM20], by computer: Γ admits no automorphism group of order 6 or 9.
- Makhnev [Mak10, Corollary 2.4]: if 2 divides $|G|$, then $|G|$ divides 42. Cesarz and Woldar [CW25], without computers: if 7 divides $|G|$, then $G \simeq \mathbb{Z}/7\mathbb{Z}$; consequently, if 2 divides $|G|$, then $|G|$ divides 6.
- Ishida [I26, Proposition 8.7], by a trace–rank identity: a fixed-point-free automorphism of order 3 has exactly six orbits that induce a triangle.

Together, these results give $|G| \in \{1, 2, 3\}$: the only remaining open case was a fixed-point-free automorphism of order 3. It is still stated as unresolved in [TS26], after an attempt with a CP-SAT solver (see also [K26] for an approach with a SAT solver without a symmetry assumption). Theorem 1 closes this case; Proposition 2 and Lemma 7 reprove the fixed-point case without computers. Corollary 8 (§5) draws the consequences for $|G|$. The argument uses the following invariant, the discriminant, and also recovers Ishida's number six (Remark 6).

2 Reduction to a statement about matrices

Let g be a fixed-point-free automorphism of order 3. Its 33 orbits each have three elements; choose a representative in each and denote the vertices by (U, b), $U \in \{1, \dots, 33\}$, $b \in \mathbb{Z}/3$, with $g(U, b) = (U, b + 1)$. For $b \in \mathbb{Z}/3$, define $A_b \in M_{33}(\mathbb{Z})$ by $A_b[U, V] = A[(U, 0), (V, b)]$. Since g is an automorphism, $A[(U, c), (V, c + b)] = A_b[U, V]$ for every c, and symmetry of A gives $A_b^{\mathsf{T}} = A_{-b}$. Set

$$Q = A_0 + A_1 + A_2 \in M_{33}(\mathbb{Z}), \qquad M = A_0 + \omega A_1 + \omega^2 A_2 \in M_{33}(\mathbb{Z}[\omega]).$$

Lemma 4. *(i) Q is symmetric, $Qj = 14j$, and $Q^2 + Q = 12I + 6J$. (ii) M is Hermitian and $M^2 + M = 12I$. (iii) $Q - M = (1 - \omega)A_1 + (1 - \omega^2)A_2$ is divisible by π entrywise. (iv)* $\operatorname{tr} Q = 2\tau$ *and* $\operatorname{tr} M = -\tau$*, where τ is the number of orbits that induce a triangle.*

Proof. For a matrix X indexed by the vertices and commuting with g, set $X_b[U, V] = X[(U, 0), (V, b)]$ and $\Phi(X) = \sum_b X_b x^b$, an element of $M_{33}(\mathbb{Z}[x]/(x^3 - 1))$. We have $(XY)_b = \sum_c X_c Y_{b-c}$, so Φ is a ring homomorphism, with $\Phi(I) = I$ and $\Phi(J) = (1 + x + x^2)J$. Apply Φ to (1) and then evaluate at $x = 1$ and $x = \omega$ (where $1 + \omega + \omega^2 = 0$) to obtain the equations in (i) and (ii). The row sums of Q are the degrees, hence $Qj = 14j$. We have $Q^{\mathsf{T}} = \sum_b A_{-b} = Q$ and $M^* = \sum_b \bar{\omega}^b A_{-b} = M$. Part (iii) follows from $1 - \omega^2 = (1 - \omega)(1 + \omega)$. For (iv), $(U, 0)$ is adjacent to $(U, 1)$ if and only if it is adjacent to $(U, 2)$, precisely for triangular orbits; hence $Q[U, U]$ is 2 or 0, and $M[U, U]$ is $\omega + \omega^2 = -1$ or 0. □

Theorem 1 therefore follows from Lemma 4 and the following statement.

Theorem 5. *There is no pair (Q, M) such that $Q \in M_{33}(\mathbb{Z})$ is symmetric with $Qj = 14j$ and $Q^2 + Q = 12I + 6J$, $M \in M_{33}(\mathbb{Z}[\omega])$ is Hermitian with $M^2 + M = 12I$, and $Q \equiv M \pmod{\pi}$ entrywise.*

3 Proof of Theorem 5

Let $\rho : \mathbb{Z}[\omega] \to \mathbb{F}_3$ be the reduction homomorphism ($\rho(\omega) = 1$, with kernel $\pi\mathbb{Z}[\omega]$). On $\mathbb{Z}$, it is reduction modulo 3, and $\rho(\overline{z}) = \rho(z)$. For a matrix X, write $\overline{X} = \rho(X)$; then $\rho(X^*) = \overline{X}^{\mathsf{T}}$.

Step 1 (identities). Let $V = \ker(Q - 3I) \subset \mathbb{Q}^{33}$. (a) If $x \in V$, then $j^{\mathsf{T}}x = 0$: indeed, $14j^{\mathsf{T}}x = j^{\mathsf{T}}Qx = 3j^{\mathsf{T}}x$. (b) The symmetric integer matrix $N = 11(Q + 4I) - 6J$ satisfies $(Q - 3I)N = 0$ and $Nx = 77x$ for $x \in V$. Indeed, $(Q - 3I)(Q + 4I) = Q^2 + Q - 12I = 6J$ and $(Q - 3I)J = 11J$, so $(Q - 3I)N = 66J - 66J = 0$; if $x \in V$, then $(Q + 4I)x = 7x$ and $Jx = 0$. (c) The Hermitian matrix $N' = M + 4I$ satisfies $(M - 3I)N' = M^2 + M - 12I = 0$ and $N'y = 7y$ for $y \in \ker(M - 3I)$. (d) Since Q is symmetric and $Qj = 14j$, we have $QJ = JQ = 14J$; on $j^{\perp}$, $Q^2 + Q = 12I$, so 14 is a simple eigenvalue and

$$W := \ker((Q - 3I)(Q - 14I)) = V \oplus \mathbb{Q}j.$$

The symmetric integer matrix $N_W = 3(Q + 4I) - J$ satisfies $(Q - 3I)(Q - 14I)N_W = 18(Q - 14I)J = 0$ and $N_W w = 21w$ for $w \in W$ (on V by (a); and $N_W j = 54j - 33j$).

Step 2 (integral bases). Let R be a principal ideal domain ($\mathbb{Z}$ or $\mathbb{Z}[\omega]$) and $X \in M_n(R)$. The module $L = \{v \in R^n : Xv = 0\}$ is a direct summand of R^n: indeed $R^n/L \simeq X(R^n) \subset R^n$ is finitely generated and torsion-free, hence free. Thus there is an $n \times r$ matrix B whose columns form a basis of L (r is the dimension of the kernel over the fraction field), and an $r \times n$ matrix C such that $CB = I_r$; then $BCv = v$ for every $v \in L$. Apply this to:

- $L = \ker_{\mathbb{Z}}(Q - 3I)$: (B, C, r), $G = B^{\mathsf{T}}B$, $d = \det G > 0$;
- $L' = \ker_{\mathbb{Z}[\omega]}(M - 3I)$: (B', C', r') and $G' = B'^{*}B'$, a positive-definite Hermitian matrix with determinant $d' = \det G' > 0$; since d' is real and belongs to $\mathbb{Z}[\omega]$, $d' \in \mathbb{Z}$;
- $L_W = \ker_{\mathbb{Z}}((Q - 3I)(Q - 14I))$, of rank $r + 1$: (B_W, C_W), $G_W = B_W^{\mathsf{T}}B_W$.

Step 3 (the identities $GK = 77I$ and $G'K' = 7I$). Set $K = CNC^{\mathsf{T}} \in M_r(\mathbb{Z})$. The columns of N lie in L (Step 1b), so $BCN = N$; since N is symmetric, $N = NC^{\mathsf{T}}B^{\mathsf{T}}$, hence $N = BKB^{\mathsf{T}}$. Also $NB = 77B$, so $B^{\mathsf{T}}N = 77B^{\mathsf{T}}$ and

$$GK = B^{\mathsf{T}}(BCN)C^{\mathsf{T}} = B^{\mathsf{T}}NC^{\mathsf{T}} = 77(CB)^{\mathsf{T}} = 77I_r.$$

Similarly, with $K' = C'N'C'^{*}$ and $K_W = C_W N_W C_W^{\mathsf{T}}$: $N' = B'K'B'^{*}$, $G'K' = 7I_{r'}$, and $G_W K_W = 21I_{r+1}$. Consequences:

- $d \det K = 77^r$, so $d = 7^a 11^b$;
- K' is Hermitian, so $\det K' \in \mathbb{Z}$, and $d' \det K' = 7^{r'}$; thus $d' = 7^c$, and $d' \equiv 1 \pmod 3$;
- $\det G_W$ divides 21^{r+1}; in particular, $11 \nmid \det G_W$.

Step 4 (the exponent b is odd). The columns of $[B \mid j]$ lie in L_W, so $[B \mid j] = B_W T$ with $T = C_W[B \mid j] \in M_{r+1}(\mathbb{Z})$. By Step 1(a), $B^{\mathsf{T}}j = 0$: the Gram matrix of $[B \mid j]$ is $\mathrm{diag}(G, 33)$ and also $T^{\mathsf{T}}G_W T$. Hence

$$33d = \det(T)^2 \det G_W.$$

The 11-adic valuations give $1 + b = 2v_{11}(\det T)$: b is odd. Thus $d = 7^a 11^b \equiv (-1)^b = -1 \pmod 3$.

Step 5 (reduction modulo 3: $d \equiv d'$). Since $Q \equiv M \pmod{\pi}$, $\bar{N} = 2(\bar{Q} + I)$ and $\bar{N}' = \bar{Q} + I$; thus $\bar{N} = 2\bar{N}'$ and $\bar{N}' = 2\bar{N}$. Reducing the identities of Step 3 gives

$$\bar{K} = \bar{C}\bar{N}\bar{C}^{\mathsf{T}} = 2(\bar{C}\bar{B}')\bar{K}'(\bar{C}\bar{B}')^{\mathsf{T}},$$
$$\bar{K}' = \bar{C}'\bar{N}'\bar{C}'^{\mathsf{T}} = 2(\bar{C}'\bar{B})\bar{K}(\bar{C}'\bar{B})^{\mathsf{T}}.$$

The determinants $\det K = 77^r/d$ and $\det K' = 7^{r'}/d'$ are coprime to 3, so $\bar{K}$ and $\bar{K}'$ are invertible. The first identity factors $\bar{K}$ through $\mathbb{F}_3^{r'}$, so $r \le r'$; the second gives $r' \le r$. Thus $r = r'$, $\bar{C}\bar{B}'$ is square and invertible, and taking determinants in the first identity yields

$$2^r\rho(d)^{-1} = \det\bar{K} = 2^r \det(\bar{C}\bar{B}')^2 \det\bar{K}' = 2^r\rho(d')^{-1},$$

since every nonzero square in $\mathbb{F}_3$ is 1, $\rho(77) = 2$, and $\rho(7) = 1$. Hence $d \equiv d' \pmod 3$.

Step 6. Steps 4 and 3 give $d \equiv 2$ and $d' \equiv 1 \pmod 3$, contradicting Step 5. □

Remark 6. $\operatorname{tr} N = \operatorname{tr}(BKB^{\mathsf{T}}) = \operatorname{tr}(KG) = 77r$, *and likewise* $\operatorname{tr} N' = 7r'$. *By Lemma 4(iv),* $77r = 11(2\tau + 132) - 198$ *and* $7r' = 132 - \tau$. *The equality* $r = r'$ *from Step 5 therefore gives* $2\tau + 114 = 132 - \tau$, *so* $\tau = 6$ *and* $r = r' = 18$*: this recovers Ishida's result by the same reduction. The new obstruction is the next step: the equality of the discriminants.*

4 The case of three fixed points (Proposition 2)

Let g have order 3 and exactly three fixed points t_1, t_2, t_3. They form a triangle: if t_i and t_j were nonadjacent, g would permute their $\mu = 2$ common neighbors, and hence fix them (since it has order 3), giving at least four fixed points. The other 96 vertices form 32 free orbits with vertices (U, b), $U \le 32$; define A_b, Q, M (of size 32×32) as in §2, and $a[U, i] = A[(U, 0), t_i]$. Then M is Hermitian, $M^2 + M = 12I$ (in the calculation of §2, paths through a t_i and the J term are multiplied by $1 + \omega + \omega^2 = 0$), and $Q \equiv M \pmod{\pi}$.

The vectors in $\mathbb{Z}^{99}$ invariant under g are identified with $\mathbb{Z}^{35}$ (one value per free orbit, one per t_i). The action of A on them is

$$\widetilde{A} = \begin{pmatrix} Q & a \\ 3a^{\mathsf{T}} & J_3 - I_3 \end{pmatrix}, \qquad \widetilde{A}j = 14j, \qquad \widetilde{A}^2 + \widetilde{A} = 12I + 2jw^{\mathsf{T}},$$

where $w = (3, \dots, 3, 1, 1, 1)$ (J acts as jw^{T}). The inner product on $\mathbb{Z}^{99}$ induces the form $D = \operatorname{diag}(w)$, with respect to which $\widetilde{A}$ is self-adjoint ($D\widetilde{A}$ is symmetric). Put $\Delta = \operatorname{diag}(I_{32}, 3I_3)$, so that $D\Delta = 3I$.

1. *The fixed-point coordinates are divisible by* 3. Let $\widetilde{N} = 11(\widetilde{A}+4I) - 2jw^{\mathsf{T}}$. As in Step 1, $(\widetilde{A}-3I)\widetilde{N} = 0$ and $\widetilde{N}x = 77x$ if $\widetilde{A}x = 3x$ (then $w^{\mathsf{T}}x = 0$). The last three rows of $\widetilde{N}$ vanish modulo 3: $\widetilde{N}[t_i, U] = 33a[U, i] - 6$ and $\widetilde{N}[t_i, t_j] = 9 + 33\delta_{ij}$. If $x \in L = \ker_{\mathbb{Z}}(\widetilde{A} - 3I)$, then $77x_{t_i} \equiv 0 \pmod 3$, so $3 \mid x_{t_i}$.

2. *Weighted Gram matrix.* Let B be a basis of L (rank r) and C satisfy $CB = I$. By item 1, $\widetilde{G} = \frac{1}{3}B^{\mathsf{T}}DB$ is an integral, positive-definite matrix. The matrix $X = \widetilde{N}\Delta$ is symmetric, $DX = 3\widetilde{N}^{\mathsf{T}}$, and its columns lie in L. As in Step 3, $X = BKB^{\mathsf{T}}$ with integral $K = CXC^{\mathsf{T}}$, and $3\widetilde{G}K = B^{\mathsf{T}}DXC^{\mathsf{T}} = 3(\widetilde{N}B)^{\mathsf{T}}C^{\mathsf{T}} = 3 \cdot 77I$. Hence $\widetilde{G}K = 77I$ and $\det\widetilde{G} = 7^a 11^b$.

3. *Reduction modulo* 3. We have $\bar{X} = \operatorname{diag}(2(\bar{Q} + I), 0)$ (the last three columns are multiplied by 3, and the last three rows vanish) and $\bar{B} = \binom{\bar{B}_1}{0}$ by item 1. Modulo 3, the identities $X = BKB^{\mathsf{T}}$ and $K = CXC^{\mathsf{T}}$ therefore involve only the 32×32 block:

$$2(\bar{Q} + I) = \bar{B}_1\bar{K}\bar{B}_1^{\mathsf{T}}, \qquad \bar{K} = 2\bar{C}_1(\bar{Q} + I)\bar{C}_1^{\mathsf{T}}.$$

On the M side, Steps 2 and 3 apply unchanged ($d' = 7^c \equiv 1 \pmod 3$), and Step 5, verbatim, gives $r = r'$ and $\det\widetilde{G} \equiv d' \equiv 1 \pmod 3$.

4. *The prime* 11. $W = \ker_{\mathbb{Z}}((\widetilde{A} - 3I)(\widetilde{A} - 14I))$ contains L and j and has rank $r + 1$ (on $\{w^{\mathsf{T}}x = 0\}$, $\widetilde{A}^2 + \widetilde{A} = 12I$). The matrix $N_W = 9(\widetilde{A} + 4I) - jw^{\mathsf{T}}$ is 63 times the projector onto $W \otimes \mathbb{Q}$; with $X_W = N_W\Delta$, the calculation in item 2 gives $G_W K_W = 189I$ for $G_W = B_W^{\mathsf{T}} D B_W$, hence $11 \nmid \det G_W$. Since $w^{\mathsf{T}}B = 0$, the Gram matrix (for D) of $[B \mid j]$ is $\operatorname{diag}(3\widetilde{G}, j^{\mathsf{T}}Dj) = \operatorname{diag}(3\widetilde{G}, 99)$; as in Step 4,

$$3^r \cdot 99 \cdot \det \widetilde{G} = \det(T)^2 \det G_W.$$

Thus b is odd and $\det \widetilde{G} \equiv 2 \pmod 3$, contradicting item 3.

5 Fixed points (Corollary 3) and the order of $\mathrm{Aut}(\Gamma)$

Lemma 7. *Let $g \neq 1$ be an automorphism of Γ with $g^3 = 1$. Then g has at most three fixed points.*

Proof. Let F be the set of fixed points and, for $x \in F$, let n_x be the number of fixed neighbors of x.

1. *Closure.* A vertex v adjacent to two distinct fixed points a, b is fixed: if $a \sim b$, v is their unique common neighbor ($\lambda = 1$); otherwise g permutes the $\mu = 2$ common neighbors of a, b and, being of order 3, fixes them. In particular, a nonfixed vertex has at most one fixed neighbor.

2. $n_x \in \{2, 8, 14\}$. The nonfixed neighbors of $x \in F$ form orbits of size 3, so $n_x \equiv 14 \equiv 2 \pmod 3$. Moreover n_x is even: by $\lambda = 1$, a fixed neighbor u of x has a unique common neighbor with x, which is fixed by item 1; this pairs the fixed neighbors of x.

3. *Counting.* By item 1, there are at most $99 - |F|$ edges between F and its complement:

$$\sum_{x \in F} (14 - n_x) \leq 99 - |F|, \qquad \text{or} \qquad 15|F| \leq 99 + \sum_{x \in F} n_x.$$

4. *Conclusion.* If $n_x = 14$, every vertex nonadjacent to x has two common neighbors with x, both fixed, and hence is fixed (item 1): $g = 1$, contrary to the assumption. If $n_x = 8$ and no n_y equals 14, let A be the set of the eight fixed neighbors of x and $S = F \setminus (\{x\} \cup A)$, of size s. A vertex $u \in A$ has as fixed neighbors x, at most one vertex of A ($\lambda = 1$), and vertices of S; a vertex $y \in S$ has at most $\mu = 2$ neighbors in A. Thus $\sum_{u \in A} n_u \leq 16 + 2s$ and $\sum_{y \in F} n_y \leq 8 + (16 + 2s) + 8s$. Since $|F| = 9 + s$, item 3 gives $15(9 + s) \leq 123 + 10s$, or $12 + 5s \leq 0$: impossible. It remains to consider the case in which all n_y equal 2. If $y \in F$ is neither x nor a neighbor of x, a common neighbor a of x and y is fixed, as is the common neighbor c of x and a (item 1); then a has three fixed neighbors x, y, c, a contradiction. Hence every fixed point is x or one of its two fixed neighbors: $|F| \leq 3$.

□

Combined with the published results recalled in §1, Corollary 3 gives the following.

Corollary 8. *Suppose Γ exists, and let $G = \mathrm{Aut}(\Gamma)$.*

(i) With [MM04] and [CW25], obtained without computers: $|G| \in \{1, 2, 7, 11\}$.

(ii) With the computer search [B09, BL11] as well: $|G| \in \{1, 2\}$, that is, G is trivial or isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

Proof. By Corollary 3, G has no element of order 3, so $3 \nmid |G|$ (Cauchy's theorem). (i) By [MM04], $|G|$ then divides $2 \cdot 7 \cdot 11$. If 7 divides $|G|$, then $G \simeq \mathbb{Z}/7\mathbb{Z}$ [CW25]. If 2 divides $|G|$, then $|G|$ divides 6 [CW25], so $|G| = 2$. Otherwise, $|G|$ divides 11. (ii) By [BL11], $|G|$ is then a power of 2; if it is not 1, it divides 6 [CW25], hence equals 2. □

An MIT PRIMES report [LM14], whose proofs we have not checked, also rules out order 11 by an elementary argument; then the set in (i) would become $\{1, 2, 7\}$. Corollary 8 is not formalized: the results of [MM04, B09, BL11, CW25] are assumed.

6 Remarks

- *Mechanism.* The two characters of C_3 see the same quadratic form $(\bar{Q} + I)$ over $\mathbb{F}_3$. Only the trivial character contains the vector j; its norm $33 = 3 \cdot 11$ introduces the prime 11, which is not a square modulo 3. Nothing can compensate for it: the only other prime involved is $7 = r - s$, which is 1 modulo 3 (it splits in $\mathbb{Z}[\omega]$).

- *What is used.* For Theorem 1: only (1) and the free action; neither the values $0, 1, 2$ of the entries of Q nor any computation. For Proposition 2: likewise (plus $\mu = 2$ for the triangle). An earlier version used Ishida's trace–rank identity to obtain $r = r'$; divisibility by 3 of the fixed coordinates (item 1 of §4) makes it unnecessary. For the corollary: additionally the counting in Lemma 7 ($\lambda = 1$, $\mu = 2$, $v = 99$); no external input.

- *Generalization.* The same proof gives a criterion for any $\mathrm{srg}(v, k, \lambda, \mu)$ with eigenvalues k, θ, θ', when $3 \nmid \theta - \theta'$ and all primes dividing $\theta - \theta'$ are 1 modulo 3 (integrality hypotheses checked case by case). For $v \leq 1300$, it also excludes fixed-point-free automorphisms of order 3 for $(105, 52, 21, 30)$, $(105, 52, 29, 22)$, $(261, 52, 11, 10)$, $(714, 248, 97, 80)$, $(1221, 610, 279, 330)$, and $(1221, 610, 329, 280)$. The existence of all these parameter sets is unknown in Brouwer's tables [Br].

- *Numerical checks.* The lemma "$d \equiv d' \pmod 3$" was tested without failure on the triangular graphs $T(n)$, $n \in \{6, 7, 9, 10, 12, 13, 15, 18, 21\}$, equipped with a free action of order 3 (one real example has $d \equiv d' \equiv 2$: $T(10)$), and on about 1,650 random invariant lattices, with an independent recomputation; the test does fail when the projector is not 3-integral. The pattern "d = (primes from $\theta - \theta'$) $\times$ (primes from $\langle j, j \rangle$)" is observed exactly on $T(15)$, $T(18)$, and $T(21)$.

- *Formal verification.* All statements except Corollary 8 have been verified in Lean 4 [dMU21] with Mathlib [Mat20] (version v4.35.0-rc2), without `sorry`, starting from Mathlib's definition of a strongly regular graph (`SimpleGraph.IsSRGWith`):

  - `C3.no_order3`: every automorphism σ with $\sigma^3 = 1$ is the identity (Corollary 3, without [MM04]);
  - `C3.no_free_order3` (Theorem 1);
  - `C3.no_triangle_order3` (Proposition 2; the triangle is proved there);
  - `C3.no_order3_fix_le3` (at most three fixed points);
  - `C3.no_pair` (Theorem 5) and `C3.Tri.no_triangle_pair` (§4), the matrix versions.

  All depend only on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. The final statement is isolated in `c3formal/Challenge.lean`, which imports only Mathlib and contains no definitions. It is written there in four forms: graph; no element of order 3 in $\mathrm{Aut}(\Gamma)$; 3 does not divide $|\mathrm{Aut}(\Gamma)|$; Boolean adjacency matrix on $\{0, \ldots, 98\}$, without a graph library. It comes with three non-vacuity tests: the same formulas, with $(9, 4, 1, 2)$ instead of $(99, 14, 1, 2)$, are proved false (the 3×3 rook graph is an $\mathrm{srg}(9, 4, 1, 2)$ with an automorphism of order 3). Lean 4.35's `lake comparator` tool checks that the proofs (`c3formal/Solution.lean`, derived from `C3.no_order3`) establish exactly these statements, with only these axioms, and that they are accepted, with all their dependencies down to Lean's core (more than 26,500 declarations), by Lean's kernel and five other checkers shipped with Lean: `leanchecker-paranoid`, `lean4lean` [Car24], `nanoda` [Nan], `con-leche`, and `con-ron`. Verdict: accepted, on the original machine and, in the tool's sandbox, on a fresh Linux machine (GitHub Actions). Thus a reviewer need only check `c3formal/Challenge.lean`. Sources: directory `c3formal/`; details and logs: `AUDIT_ENONCE.md`.

- *Priority for review (paper proof).* Step 2 (direct summand over $\mathbb{Z}[\omega]$), Step 4 (11-adic valuation), Step 5 (reduction), item 1 of §4, and Lemma 7. Corollary 8 also relies on [MM04, B09, BL11, CW25], which we have not rechecked.

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